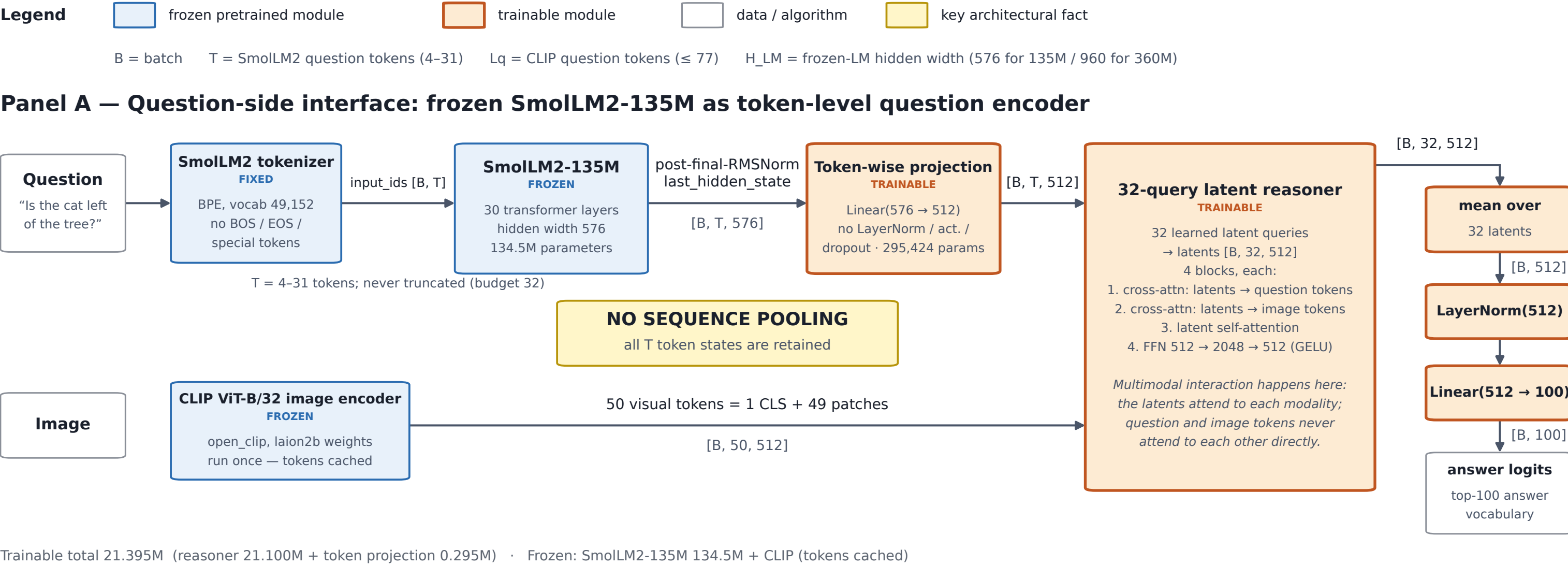
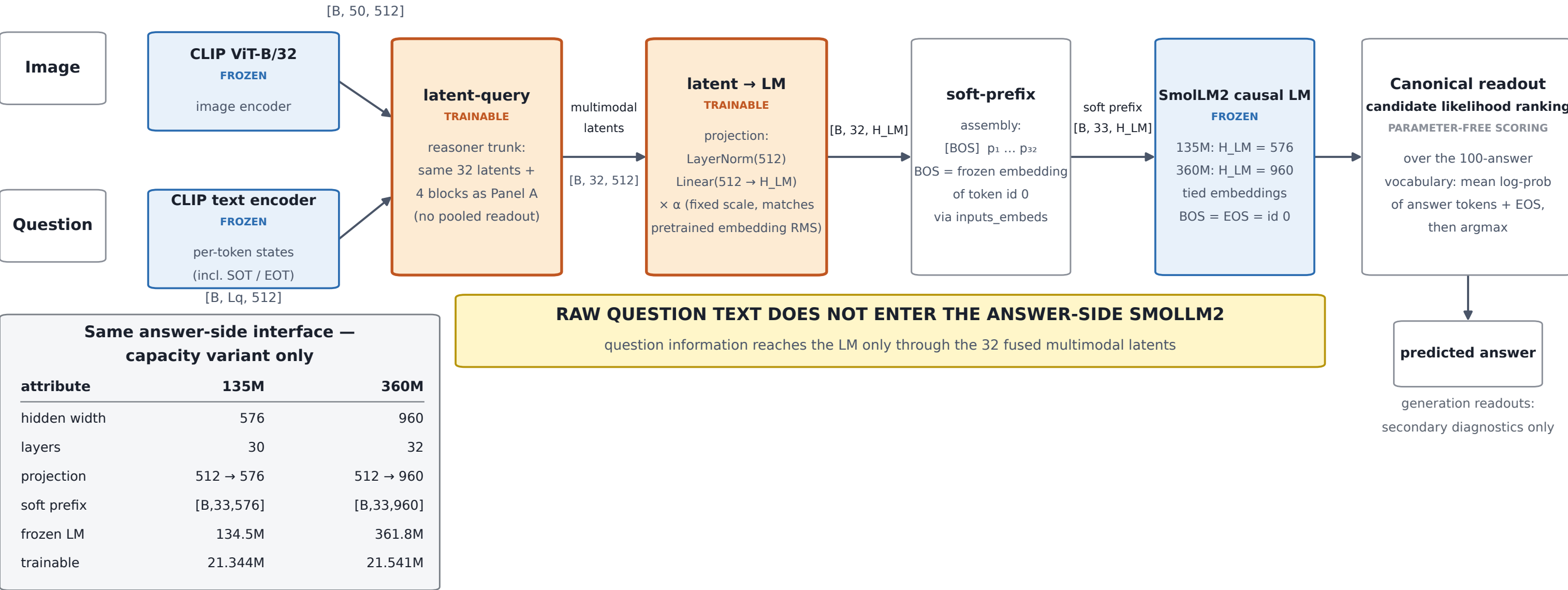


SmolLM2 interfaces in the lightweight VQA pipeline

Two placements of a frozen small language model: token-level question encoder (Panel A) and answer readout over fused multimodal latents (Panel B).
Blue components are frozen; only orange components are trained.



Panel B — Answer-side interface: frozen SmolLM2 as answer readout over fused multimodal latents



Notes

Question-side states are extracted once per unique question from the frozen LM and cached; the LM is not resident during head training.
Disclosure: CLIP image tokens and SmolLM2 question tokens originate from different pretrained spaces — the trainable projection maps into a learned common width, not a naturally shared pretrained embedding space.
Answer-side training: teacher-forced NLL of the gold answer tokens + EOS through the frozen LM; no LM weight updates. Constrained and bounded free decoding: secondary diagnostics (agree within ≈ 0.0005).
Control: a language-model-free attention-pool readout over the same 32 latents (52.8k parameters).
Internal experiment IDs — question side: E8A (A1 pretrained, A1r random-init, A0p CLIP-token control); answer side: E8B (B3 pretrained / B2 random-init, B1 LM-free control at 135M) and E10 (B4 / B4r at 360M).